

claude-windows / 6a82308f-69d6-43a8-9ea4-d8eaf497253d

Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\subagents\workflows\wf_3d536535-241\agent-a7ec470e04fb42857.jsonl`  
- SHA-256: `23031de20c8f96951ca71f0920471afb56f3e6e5a87e9a82a0379e9efcc43089`  
- Source modified: `2026-06-16T03:21:35+00:00`  
- Imported at: `2026-07-05T16:48:27+00:00`  
- project: `wf_3d536535-241`  
- session_id: `6a82308f-69d6-43a8-9ea4-d8eaf497253d`
```

Transcript

1. user (2026-06-16T03:20:14.238Z)

Source Extractor

Research question: "Research question: **What concrete, recent (2022?2025) techniques and evaluation practices should a local, commercially-licensed badminton rally-detector adopt to close the quality gap with a strong cloud VLM (Gemini), given a tiny labeled set (~6 ? ~16 golden videos)?** Produce an adversarially-verified, cited report organized around the four dimensions below. Prefer specific named methods, papers (arXiv/venue), and reported numbers over generic advice; flag anything you cannot verify.

CONTEXT (so you target the real gap, not generic ML):

- System: a local pipeline detects the shuttle per-frame (WASB / TrackNet, MIT-licensed) ? a heuristic turns the trajectory into candidate rally windows ? today an optional Gemini "handoff" confirms rallies. We just measured a fully-local (no-AI) run vs Gemini on 3 matches: local emits far FEWER but MUCH LONGER windows (mean 65s vs 6.1s; one window spanned a whole 174s clip) ? it does NOT miss activity, it FAILS TO SEGMENT. Root cause (code-confirmed): a frame is "active" iff the shuttle is visible AND moving above a small per-second-normalized velocity threshold; active frames within a 2 s gap are merged; there is NO max-window cap and NO inactivity/boundary split. So "shuttle is moving" is conflated with "rally in progress," and dead-time + serves + knock-ups bridge rallies into mega-windows. Shuttle tracking itself is excellent (96?99% per-frame visibility).

- HARD CONSTRAINTS: weights/data/code must be MIT/Apache-2.0/BSD only (no viral/NC/custom licenses, reject Gemma-3/Llama-4-MAU-cap); we may NEVER train owned weights on paid-LLM (Gemini/OpenAI/Anthropic) outputs (terms/copyright), though paid LLMs may be used at inference; minors excluded from any training pool; runs on a single local GPU.

- WHAT WE ALREADY HAVE (do NOT re-survey the basics ? go deeper/newer): the architecture skeleton is known to be standard Temporal Action Localization/Segmentation (proposal+scorer) + late/score-level fusion + stacked generalization; in-domain prior art (MonoTrack, ShuttleSet, See et al. 2024/25 non-broadcast badminton rally start/end, Ghosh et al. point-segmentation, tennis/TT play-break state machines) is already cited. We have a working eval harness: leave-one-video-out (grouped by match), temporal-IoU F1 @0.5, F1@{0.1,0.25,0.5}, mAP@tIoU, segment-count-ratio, bootstrap 95% CIs, paired exact sign-test, Platt/isotonic calibration + Brier/ECE, a tiny numpy LogisticRegression scorer over per-window feature columns, an experiment leaderboard, and an "eval?serve" guard (a cue measured +0.074 F1 on permissive proposal windowing but ?0.008 on the coarser served windowing ? reverted). Available but default-OFF per-window signals: net-crossing count (rally gate), 5 person/court features, 3 audio-hit features.

THE FOUR DIMENSIONS:

1) WINDOWING & BOUNDARIES (Temporal Action Localization/Segmentation): beyond proposal+score ? what are the best *recent* methods for (a) turning a dense per-frame signal into well-segmented actions, (b) explicitly controlling OVER-segmentation/merging, (c) boundary detection/regression and splitting merged segments? Cover e.g. MS-TCN/MS-TCN++, ASRF (action-boundary regression), BMM/BSN boundary-matching, ActionFormer/TriDet/TemporalMaxer (actionness + boundary heads), smoothing/boundary losses, and the standard metrics for over-segmentation (segmental F1@k,

edit/Levenshtein score, mAP@tIoU). Which transfer to a 1-D shuttle-trajectory + auxiliary-cue setting on tiny data?

2) SHUTTLE-TRAJECTORY ? RALLY SIGNAL PROCESSING (racket-sports specific): how do published systems convert ball/shuttle tracking into rally/point segments and distinguish "in-play" from dead-time? Cover rally-state machines, serve/let detection, net-crossing & two-sided-occupancy cues, hit/contact (audio + kinematic) detection, and fps-invariant trajectory features. What features best separate "rally" from "shuttle merely moving"?

3) SMALL-DATA STRATEGIES: with ~6716 labeled videos, what gives the most quality per label ? active learning (which videos/segments to label next; uncertainty/diversity/core-set), semi-/weakly-/point-supervised TAL (e.g. SF-Net, timestamp supervision), self-training/pseudo-labeling, and data-efficient TAL? Separately: the legally-clean ways to exploit a strong teacher (Gemini) at INFERENCE for weak/pseudo-labels or as an evaluation oracle WITHOUT its outputs becoming training data for owned weights ? what does the literature say about teacher-student / distillation boundaries and label-noise handling, and what governance distinctions matter?

4) EVALUATION METHODOLOGY & RIGOR at small n: trustworthy measurement with ~6716 videos ? bootstrap/permutation tests, paired sign-tests, leave-one-group-out vs RL00CV bias, calibration (reliability, ECE), the over-segmentation-blindness of plain tIoU, shadow/A-B evaluation, regression gates, and concrete pitfalls (label noise, annotator agreement/IAA, multiple-comparison/over-fitting traps, eval-vs-serve drift). What metrics + protocols should a small-data sports-segmentation leaderboard standardize on?

Deliverable: a structured, cited report with, per dimension, the 375 highest-leverage techniques that APPLY to our constrained setting (small data, local, license-clean, 1-D trajectory + cues + learned scorer), each with: what it is, why it fits our over-merge/small-data problem, the expected difficulty/payoff, and the citation. Call out which ideas are low-effort high-confidence wins vs research bets. Be explicit about anything unverified or license-incompatible."

Fetch and extract key claims from this source:

****URL:**** https://link.springer.com/chapter/10.1007/978-3-031-69073-0_15

****Title:**** Service and End of Rally Detection in Badminton Videos (See, Paramesran, Krishnasamy, 2024)

****Found via:**** Shuttle/ball trajectory ? rally/point segmentation in racket sports search

Task

1. Use WebFetch to retrieve the page content.
2. Assess source quality: primary research/institution? secondary reporting? blog/opinion? forum? unreliable?
3. Extract 2-5 FALSIFIABLE claims that bear on the research question. Each claim must:
 - be a concrete, checkable statement (not vague generalities)
 - include a direct quote from the source as support
 - be rated central/supporting/tangential to the research question
4. Note publish date if available.

If the fetch fails or the page is irrelevant/paywalled, return claims: [] and sourceQuality: "unreliable".

Structured output only.

2. assistant (2026-06-16T03:20:16.484Z)

I'll fetch the source page first.

3. user (2026-06-16T03:20:23.948Z)

REDIRECT DETECTED: The URL redirects to a different host.

Original URL: https://link.springer.com/chapter/10.1007/978-3-031-69073-0_15

Redirect URL: https://idp.springer.com/authorize?response_type=cookie&client_id=springerlink&redirect_uri=https%3A%2F%2Flink.springer.com%2Fchapter%2F10.1007%2F978-3-031-69073-0_15

Status: 303 See Other

To complete your request, I need to fetch content from the redirected URL. Please use WebFetch again with these parameters:

- url: "https://idp.springer.com/authorize?response_type=cookie&client_id=springerlink&redirect_uri=https%3A%2F%2Flink.springer.com%2Fchapter%2F10.1007%2F978-3-031-69073-0_15"
- prompt: "Extract key technical claims about: (1) how shuttle/ball trajectory is converted into rally/point segments, (2) service detection and end-of-rally detection methods, (3) any reported accuracy/precision/recall/F1 numbers, (4) the dataset used (number of videos, broadcast vs non-broadcast), (5) features used to distinguish in-play from dead-time, (6) any methods for segmenting/splitting merged rallies. Provide direct verbatim quotes for each claim. Also note the publication date/year and venue."

4. assistant (2026-06-16T03:20:27.960Z)

The Springer page redirects to an authentication gate. Let me try alternate sources for this paper ? the publisher abstract page and a search for an open-access version.

5. user (2026-06-16T03:20:36.971Z)

Web search results for query: "See Paramesran Krishnasamy "Service and End of Rally Detection in Badminton Videos" 2024"

Links: [{"title": "dblp: Ganesh Krishnasamy", "url": "https://dblp.org/pid/147/9353.html"}, {"title": "Service and End of Rally Detection in Badminton Videos | Springer Nature Link", "url": "https://link.springer.com/chapter/10.1007/978-3-031-69073-0_15"}, {"title": "A Comparative Analysis of Deep Learning Models and Gradient Computation for Rally Detection in Badminton Videos | SN Computer Science | Springer Nature Link", "url": "https://link.springer.com/article/10.1007/s42979-025-03935-0"}, {"title": "A Comparative Analysis of Deep Learning Models and Gradient Computation for Rally Detection in Badminton Videos - Monash University", "url": "https://research.monash.edu/en/publications/a-comparative-analysis-of-deep-learning-models-and-gradient-compu"}, {"title": "Ready-to-Serve Detection in Badminton Videos | Semantic Scholar", "url": "https://www.semanticscholar.org/paper/Ready-to-Serve-Detection-in-Badminton-Videos-Yue-Paramesran/0744a83d324e9b2d87577ebd5ec5f238744fceb5"}, {"title": "Sivarajan Solaimalai", "url": "https://en.wikipedia.org/wiki/Sivarajan_Solaimalai"}]

I found the paper you're looking for! Here are the details:

"Service and End of Rally Detection in Badminton Videos" is a 2024 paper by See Shin Yue, Mohd Fikree Hassan, Ganesh Krishnasamy, Junn Yong Loo, and Raveendran Paramesran, published at ISACE 2024 in the Lecture Notes in Computer Science volume 14794.

The paper presents a technique to detect the start of service and end of rally, utilizing court details and poses to accurately identify the service action from other badminton actions. The research addresses the need to eliminate manual annotation, which is time-consuming for coaches and players analyzing badminton match videos.

You can access the paper through the Springer Nature Link:
https://link.springer.com/chapter/10.1007/978-3-031-69073-0_15

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. assistant (2026-06-16T03:20:39.337Z)

Let me try to get more detail from the DBLP entry and a possible open-access version, plus the related Monash publication page which may carry the abstract.

7. user (2026-06-16T03:20:43.012Z)

REDIRECT DETECTED: The URL redirects to a different host.

Original URL: https://idp.springer.com/transit?redirect_uri=https%3A%2F%2Flink.springer.com%2Fch

apter%2F10.1007%2F978-3-031-69073-0_15&code=19902d53-8806-4e8a-99e3-7374428563c4

Redirect URL: [https://link.springer.com/chapter/10.1007/978-3-031-69073-](https://link.springer.com/chapter/10.1007/978-3-031-69073-0_15?error=cookies_not_supported&code=19902d53-8806-4e8a-99e3-7374428563c4)

[0_15?error=cookies_not_supported&code=19902d53-8806-4e8a-99e3-7374428563c4](https://link.springer.com/chapter/10.1007/978-3-031-69073-0_15?error=cookies_not_supported&code=19902d53-8806-4e8a-99e3-7374428563c4)

Status: 302 Found

To complete your request, I need to fetch content from the redirected URL. Please use WebFetch again with these parameters:

- url: "https://link.springer.com/chapter/10.1007/978-3-031-69073-

[0_15?error=cookies_not_supported&code=19902d53-8806-4e8a-99e3-7374428563c4](https://link.springer.com/chapter/10.1007/978-3-031-69073-0_15?error=cookies_not_supported&code=19902d53-8806-4e8a-99e3-7374428563c4)"

- prompt: "Extract the abstract and any technical claims about: service detection method, end-of-rally detection method, court detection and pose features used, reported accuracy/precision/recall/F1 numbers, dataset size and whether broadcast or non-broadcast video. Provide verbatim quotes. Note publication year and venue."

8. user (2026-06-16T03:20:49.566Z)

Web search results for query: "Service and End of Rally Detection in Badminton Videos" abstract pose court accuracy ISACE 2024 See"

Links: [{"title": "Service and End of Rally Detection in Badminton Videos | Springer Nature Link", "url": "https://link.springer.com/chapter/10.1007/978-3-031-69073-0_15"}, {"title": "A Comparative Analysis of Deep Learning Models and Gradient Computation for Rally Detection in Badminton Videos | SN Computer Science | Springer Nature Link", "url": "https://link.springer.com/article/10.1007/s42979-025-03935-0"}, {"title": "Detection of service activity in a badminton game | IEEE Conference Publication | IEEE Xplore", "url": "https://ieeexplore.ieee.org/document/6129115/"}, {"title": "Automated service scene detection for Badminton game analysis Using CHLAC and MRA | Request PDF", "url": "https://www.researchgate.net/publication/290557755_Automated_service_scene_detection_for_Badminton_game_analysis_Using_CHLAC_and_MRA"}, {"title": "Towards Structured Analysis of Broadcast Badminton Videos", "url": "https://arxiv.org/pdf/1712.08714"}, {"title": "Automatic rally detection and scoring", "url": "https://image-pubs.uspto.gov/dirsearch-public/print/downloadPdf/10751601"}, {"title": "MonoTrack: Shuttle trajectory reconstruction from monocular badminton video", "url": "https://arxiv.org/pdf/2204.01899"}, {"title": "BST: Badminton Stroke-type Transformer for Skeleton-based Action Recognition in Racket Sports", "url": "https://arxiv.org/html/2502.21085v1"}, {"title": "Best of badminton long rallies at the 2024 Paris Olympics ...", "url": "https://m.youtube.com/watch?v=JJ7PDh_Q6Ws"}]

I found the paper you're looking for! The paper "Service and End of Rally Detection in Badminton Videos" was authored by See, S.Y., Hassan, M.F., Krishnasamy, G., Loo, J.Y., and Paramesran, R. (2024), and it was published in ISACE 2024 as part of the Lecture Notes in Computer Science series, volume 14794.

Key Details about the Research:

For detecting the start of service, the researchers used two different models: Bidirectional Long Short Term Memory (BiLSTM) and Convolutional Neural Network (CNN), while gradient computation determined the end of the rally.

A total of 12 frames with 26 features from each frame comprising distance, angle, gradient, and movements typical of serving and receiving actions were arranged in a 2D format and used as inputs to the models.

A comparative analysis of both models' classification performance was studied to identify the frame interval separation that gave the best classification accuracy.

The paper is available through Springer Nature Link at:
https://link.springer.com/chapter/10.1007/978-3-031-69073-0_15

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

9. assistant (2026-06-16T03:20:56.513Z)

The Springer page is behind an auth/cookie gate. Let me follow the cookie-not-supported redirect

to attempt the abstract, and also fetch the closely-related follow-on journal article (which extends this work and is more accessible) plus the Monash research page for corroborating detail.

10. user (2026-06-16T03:21:07.471Z)

Rally Detection in Badminton Videos - Research Summary

Publication Year: 2025

Service Detection

BiLSTM vs CNN Performance:

- BiLSTM: "highest classification accuracy of approximately 95% for the BiLSTM model, with frame intervals of 4"
- CNN: "CNN achieved similar accuracy at frame intervals of 5"

End-of-Rally Detection

The researchers employed gradient computation methodology: "at least 80% of consecutive frames exhibit no motion of the shuttlecock to signify the end of the rally" with "average accuracy of 95% across a total of 187 rallies"

Feature Representation

The system used "12 frames with 26 features from each frame comprising the distance, angle, gradient and movements that are typical of a serving and receiving actions was arranged in a 2D format"

Dataset Composition

- **Training set:** "14,688 video frames, evenly divided between serving and non-serving poses, from 30 players, with 80% used for training and 20% for testing"
- **Validation set:** "4500 video frames from 10 players that are not used in the training of the models to evaluate the robustness"

Overall Accuracy Claims

Both models achieved approximately 95% classification accuracy under optimized frame interval conditions across both datasets.

11. user (2026-06-16T03:21:10.568Z)

Service and End of Rally Detection in Badminton Videos - Technical Extraction

Publication: 2024, ISACE 2024 (International Sports Analytics Conference and Exhibition)

Venue: Springer Nature, Lecture Notes in Computer Science Vol. 14794

Abstract

"Video analysis of points won and lost in a badminton match has become a valuable tool for coaches and players to analyze their matches."

Service Detection Method

The paper does **not** use BiLSTM. Instead: "The characteristics of service action are utilized to obtain the features and arranged in a 2D format for a convolutional neural network (CNN) to detect the service and non-service."

End-of-Rally Detection

"Gradients are used to detect the flight of the shuttlecock till it eventually comes to rest"

which signifies the end of the rally."

Input Representation

The paper does not explicitly specify a 12-frame or 26-feature input format in the abstract.

Court Detection and Pose Features

"It utilizes court details and poses to accurately identify the service action from the other badminton actions."

Reported Performance Metrics

"The experimental results show 88.6% accuracy for a total of 35 rally points."

No precision, recall, or F1 scores reported in the abstract.

Dataset

"Two videos of 4 and 12 min comprising 12 and 23 consecutive points are used."

Broadcast status: Not specified in provided content.

12. assistant (2026-06-16T03:21:13.990Z)

I now have solid extraction from both the original 2024 chapter abstract and its 2025 follow-on journal article. Both are primary peer-reviewed research from the same research group (See et al., Monash). I have enough to extract falsifiable claims bearing directly on dimensions 1 (windowing/boundaries) and 2 (shuttle-trajectory ? rally signal).

13. user (2026-06-16T03:21:24.471Z)

Structured output provided successfully

14. assistant (2026-06-16T03:21:35.500Z)

Extraction complete. Summary of what I retrieved and verified:

The Springer chapter page (https://link.springer.com/chapter/10.1007/978-3-031-69073-0_15) is behind a cookie/auth gate, so I extracted the abstract via the cookies-not-supported redirect and corroborated technical detail against the same group's 2025 follow-on journal article (<https://research.monash.edu/en/publications/a-comparative-analysis-of-deep-learning-models-and-gradient-compu>). Both are primary peer-reviewed research from See et al. (Monash), rated `primary`.

Key falsifiable claims extracted, all bearing on Dimensions 1 (windowing/boundaries) and 2 (shuttle-trajectory ? rally signal):

- End-of-rally is detected via shuttle gradient/flight-to-rest, operationalized as ">=80% of consecutive frames with no shuttle motion" (~95% over 187 rallies) ? a directly applicable inactivity/boundary-split rule the local over-merge problem currently lacks.
- Service (rally start) is a learned classifier over court + pose features, not a velocity heuristic.
- Compact engineered representation: 12 frames x 26 features (distance, angle, gradient, movement) in 2D ? fits a small-data, license-clean per-window scorer.
- Original 2024 result: 88.6% accuracy on just 2 videos / 35 rally points ? a published small-data precedent comparable to the 6?16 golden-video setting.

One caveat flagged for the synthesizer: the gated abstract states the 2024 chapter used a **CNN** for service detection ("convolutional neural network (CNN) to detect the service"), while the 2025 follow-on compares **BiLSTM** vs **CNN** (BiLSTM ~95% at frame-interval 4, CNN similar at interval 5). The "26-feature / 12-frame" and "80%-no-motion" specifics come from the 2025 article; treat them as the extended-work numbers, not necessarily verbatim in the 2024 chapter body (which I could not access past the abstract). Licensing of any weights/data was not stated in either accessible page.